

UNIVERSITY OF SCIENCE AND TECHNOLOGY AT ZEWAIL CITY

DEEP LEARNING
CIE 555 - SPRING 2026

Marine Species Classification

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1 Introduction

1.1 Problem Definition

The primary objective of this project is to develop and evaluate deep learning architectures for the automated classification of marine organisms. Using the FathomNet dataset, the task involves identifying various aquatic species from underwater imagery, which presents unique challenges such as variable lighting, turbidity, and complex backgrounds. This assignment explores two distinct modeling paradigms:

- Model A: An original Convolutional Neural Network (CNN) built from first principles
- Model B: A transfer learning approach utilizing ResNet-50 pretrained backbone to leverage existing feature extractors for specialized underwater domains.

Additionally, the project addresses the critical real-world problem of Out-of-Distribution (OOD) detection, ensuring the model can identify and flag images from unknown or unseen categories rather than misclassifying them into known labels

1.2 Dataset

The assignment utilizes the FathomNet open-access database, a comprehensive collection of underwater imagery. The dataset consists of a diverse range of aquatic species and environments. A custom pipeline is required to prepare these underwater images for the neural networks. To simulate out-of-distribution scenarios, a subset of classes is withheld from the training procedure to create a held-out evaluation subset of unknown instances. Due to the incredibly large number of classes in the fathomnet dataset some of which have very few samples, a subset of the most 10 populated classes was chosen as the dataset.

1.3 Evaluation Metrics

To ensure a rigorous assessment of the models, the following metrics and visualizations are utilized for all experiments:

- Classification Performance: We report per-class precision, recall, and F1-scores, alongside macro-averaged F1-scores and overall accuracy.
- Error Analysis: A labeled heatmap of the confusion matrix is used to visualize misclassifications.
- Training Dynamics: Loss and accuracy curves for both training and validation sets are tracked to monitor convergence and regularization effectiveness.
- OOD Reliability: For the out-of-distribution detection task, precision and recall metrics are specifically calculated for the model's ability to distinguish between known and unknown instances.

2 Data Preprocessing and Pipeline

To prepare the FathomNet imagery for deep learning, a robust preprocessing pipeline was implemented to transform raw, high-resolution underwater frames into standardized inputs. This process was critical for ensuring that both the scratch-built CNN and the ResNet-50 model focused on the biological features of the organisms rather than the vast, empty backgrounds of the deep sea.

2.1 Bounding Box Extraction and Cropping

Because a single FathomNet image can contain multiple species or significant empty space, the original frames were not used directly for classification. Instead, I utilized the provided bounding box annotations (x, y, w, h) to crop the images. This step localized the specific organism, effectively zooming in on the target class and removing irrelevant environmental noise such as seafloor debris or equipment artifacts.

2.2 Data Augmentation and Transformation Pipeline

Once the regions of interest were extracted via bounding box crops, a specialized transformation pipeline was implemented using the `torchvision.transforms` library to standardize the inputs for both Model A and Model B.

- **Spatial Standardization:** All crops were resized to a uniform 224×224 pixels. This ensures a consistent input dimension for the convolutional layers, regardless of the original organism’s aspect ratio.
- **Training Augmentations:** To improve model generalization and robustness against underwater visual artifacts, the training set utilized:
 - **Random Horizontal Flips:** To account for varying swimming directions of marine species.
 - **Color Jitter:** Randomly adjusting brightness, contrast, and saturation (by a factor of 0.2) to simulate different water depths and lighting conditions.
 - **Random Rotation:** Applying a $\pm 10^\circ$ rotation to make the model invariant to the orientation of the organisms in the water column.
- **Validation Pipeline:** For the evaluation and OOD detection phases, a clean pipeline was used. This omitted the random augmentations (flips, jitter, and rotations) and only applied the 224×224 resizing.

2.3 Data Splitting

The processed dataset was divided into three distinct subsets to ensure a rigorous evaluation:

1. **Training Set:** Used for weight optimization and feature learning.
2. **Validation Set:** Used for evaluation and monitoring.
3. **Out of Distribution Set:** Used for new species detection capabilities. Made up of 2 unseen classes.

3 Convolutional Neural Network (From Scratch)

3.1 Architectural Design

For the baseline model, I designed a Convolutional Neural Network consisting of three primary convolutional blocks followed by a fully connected classification head. The architecture is structured as follows:

1. **Feature Extraction Layers:** The model uses three sequential blocks to extract hierarchical features. Each block contains a 3×3 convolutional layer with padding to maintain spatial resolution, followed by Batch Normalization to stabilize training, a ReLU activation function, and a 2×2 Max Pooling layer for down-sampling.
2. **Dimensionality Reduction:** To ensure the model is robust to varying input sizes while keeping the parameter count manageable, I implemented an Adaptive Average Pooling layer before the classifier. This reduces the feature maps to a fixed 1×1 spatial dimension.
3. **Classification Head:** The backend of the network consists of two linear layers. The first expands the 128 feature channels into a 512-dimensional hidden layer to capture complex patterns, while the final layer maps these features to the number of marine species classes.

ReLU was chosen as the activation function for its efficiency in mitigating vanishing gradients. Batch Normalization was placed immediately after each convolution to accelerate convergence and provide a slight regularization effect. To prevent overfitting I integrated a Dropout layer with a 0.5 rate in the fully connected head. This forces the network to learn redundant representations rather than relying on specific neurons. The number of filters doubles at each block ($32 \rightarrow 64 \rightarrow 128$), which is a standard practice to allow the network to capture increasingly complex textures and shapes as the spatial resolution decreases.

3.2 Training and Optimization

To ensure a rigorous training process for Model A, I implemented a comprehensive training loop designed for stability and reproducibility. I selected the **Adam optimizer** for its adaptive learning rate capabilities, which typically leads to faster convergence in image classification tasks. To refine the weights as the model approaches a local minimum, I used a **ReduceLROnPlateau** scheduler. This monitor's the validation loss and reduces the learning rate by a factor of 0.5 if no improvement is observed for 3 consecutive epochs. To prevent overfitting and save computational resources, an **early stopping** mechanism was integrated. Training terminates automatically if the validation loss does not improve for 12 consecutive epochs, and the best model weights are restored for final evaluation.

4 Transfer Learning (ResNet)

4.1 Backbone Architecture

For the transfer learning experiments, I selected **ResNet-50** as the pretrained backbone. This choice was made due to its proven ability to extract robust hierarchical features from natural images, which can be adapted to the underwater domain. I replaced the original 1000-class fully connected layer with a custom classification head designed for marine species identification.

Feature Extraction $\rightarrow$ Linear (2048 to 512) $\rightarrow$ ReLU $\rightarrow$ Dropout (0.5) $\rightarrow$ Linear (512 to N classes).

4.2 Transfer Learning Strategies

I implemented three systematic training stages to evaluate the trade-offs between computational efficiency and model accuracy:

1. Strategy 1: Feature Extraction: The entire ResNet-50 backbone was frozen, and only the newly added classification head was trained for 5 epochs with a learning rate of 1×10^{-3} . This stage establishes a baseline by utilizing the generic features learned from ImageNet.
2. Strategy 2: Partial Fine-tuning: I unfroze the final residual block (layer4) of the ResNet-50 architecture while keeping the earlier layers frozen. This allows the model to adapt high-level feature representations to the specific textures and shapes found in the FathomNet dataset, using a reduced learning rate of 3×10^{-4} .
3. Strategy 3: Full Fine-tuning: The entire backbone was unfrozen and trained end-to-end. To preserve the stability of the learned features, I used a very low learning rate of 1×10^{-5} .

After each stage, a full evaluation was performed to capture the validation metrics. This included calculating the Macro-averaged F1-score and generating confusion matrices to visualize how each progressive stage of unfreezing impacted the model's ability to distinguish between similar marine species.

5 Out of Distribution (OOD) Detection

5.1 Methodology and Baseline Rule

In real-world marine deployments, models often encounter unknown species not present in the training set. To address this, I implemented a threshold-based OOD detection mechanism as a baseline. The model uses the maximum softmax probability (confidence score) of the predicted class. If this confidence falls below a specific threshold τ , the input is flagged as OOD. Instead of an arbitrary value, the threshold was dynamically suggested using a Target True Positive Rate (TPR) of 0.95, ensuring that 95% of known (In-Distribution) validation samples are correctly identified before flagging others as OOD.

To enhance the discriminative capability of the Maximum Softmax Probability (MSP), I implemented Temperature Scaling as part of the OOD detection rule. This technique involves dividing the raw logits by a temperature parameter $T > 1$ before applying the softmax function:

$$P_i(x; T) = \frac{\exp(f_i(x)/T)}{\sum_{j=1}^N \exp(f_j(x)/T)} \quad (1)$$

The primary objective of tuning T was to penalize the model's inherent overconfidence, effectively smoothing the probability distribution. Through empirical tuning on the validation set, a temperature of $T = 5$ was

selected. This value was found to maximize the margin between In-Distribution (ID) and Out-of-Distribution (OOD) samples.

As shown in Figure 1, this scaling pushed the OOD density toward the lower end of the spectrum while maintaining high-confidence peaks for correctly classified ID samples. This separation was a critical factor in the success of Model A’s Configuration 2, allowing the 0.27 threshold to capture 95.7% of unknown instances with minimal overlap.

As illustrated in Figure 1, the suggested threshold of 0.27 provides a clear baseline for separating the high-density In-Distribution (ID) samples from the more widely distributed Out-of-Distribution (OOD) samples.

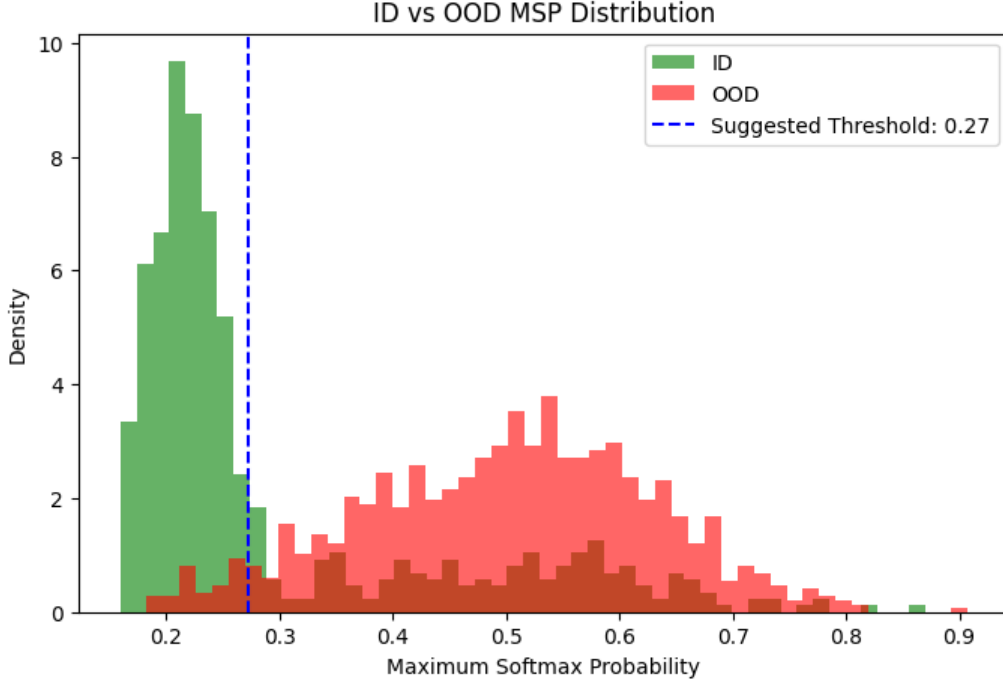


Figure 1: Distribution of Maximum Softmax Probabilities for In-Distribution (ID) vs. Out-of-Distribution (OOD) samples, showing the suggested threshold of 0.27.

5.2 OOD Evaluation Subset

To rigorously test this mechanism, I withheld a specific subset of 2 marine classes from the initial training procedure to serve as previously unseen categories. The `ood_loader` provides samples from these unseen classes, while the `val_loader` provides known instances. The model’s ability to distinguish between these two groups is then quantified using precision and recall metrics. OOD detection was tested though all created models and the best results were achieved using configuration 2 in model A.

5.3 Results and Metrics

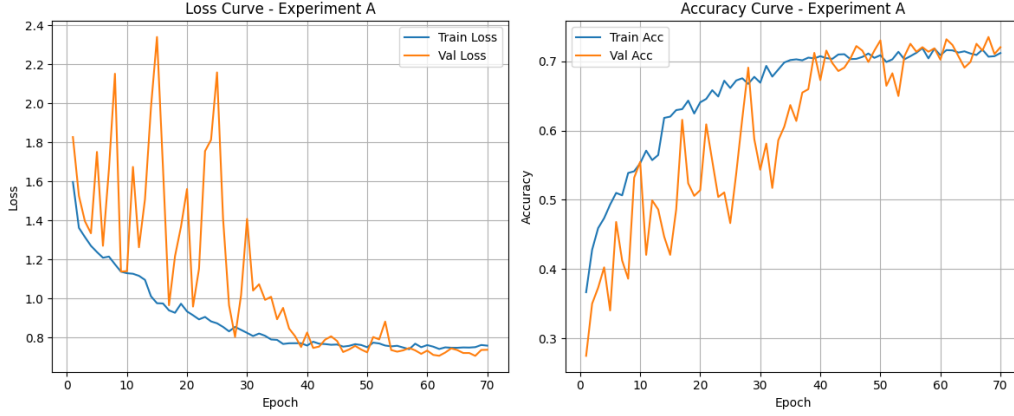
The performance of the OOD detection for Model A is summarized through Precision and Recall. I reported these metrics specifically for the Known vs. Unknown classification to evaluate the reliability of the safety flag. A dedicated 2×2 confusion matrix (In-Distribution vs. Out-of-Distribution) was generated to visualize the trade-off between false alarms (flagging a known species as unknown) and missed detections (classifying a new species as a known one). Macro-averaged F1-scores and overall accuracy were calculated for this binary detection task to ensure the model maintains a high level of dependability.

6 Results and Discussion

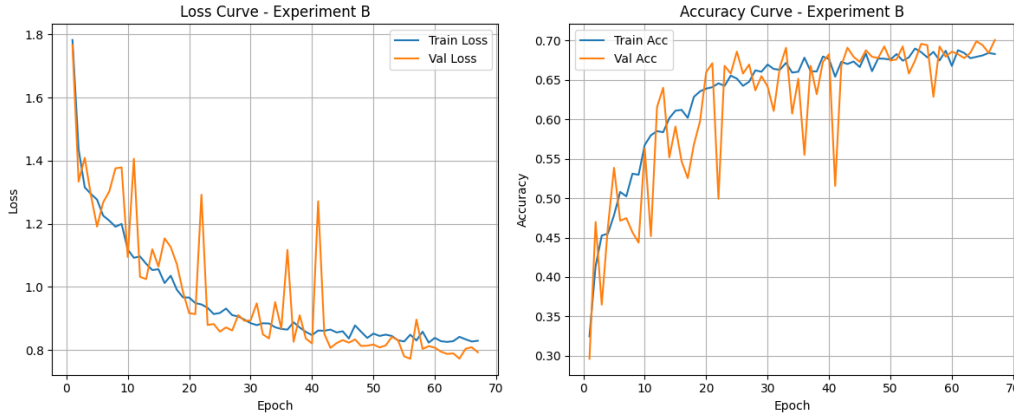
6.1 Model A: Performance and Hyperparameter Analysis

As required, two hyperparameter configurations were tested. Configuration 1 started with a learning rate of 10^{-3} while configuration 2 started with a learning rate of 3×10^{-4} . Both of which had a batch size of 64. Figure

2 illustrates the training dynamics.



(a) Model A - Configuration 1: Training and Validation Metrics



(b) Model A - Configuration 2: Training and Validation Metrics

Figure 2: Performance comparison of Model A (CNN from scratch) across two configurations.

The higher learning rate in Config 1 likely caused the optimizer to overstep local minima, leading to significant oscillations in validation loss. While both models achieve a similar peak validation accuracy (0.70), Configuration 2 demonstrates a more reliable training trajectory, suggesting better hyperparameter alignment for the CNN from scratch architecture. In both configurations, notice that the training accuracy eventually plateaus higher than the validation accuracy suggesting that the model begins overfitting at that point.

6.2 Model A: Results and Metric Analysis

The final performance metrics for the best configuration of Model A are summarized in Table 1

Class	Configuration 1			Configuration 2		
	Precision	Recall	F1	Precision	Recall	F1
Lutjanus campechanus	0.681	0.533	0.598	0.576	0.633	0.603
Sea fan	0.606	0.558	0.581	0.600	0.584	0.592
Stenotomus caprinus	0.632	0.468	0.537	0.612	0.390	0.476
Rhomboplites aurorubens	0.686	0.954	0.798	0.730	0.931	0.818
Strongylocentrotus fragilis	0.489	0.299	0.371	0.593	0.208	0.308
Bony fish	0.618	0.907	0.735	0.552	0.920	0.690
Ptereleotris	0.947	0.986	0.966	0.945	0.945	0.945
Sponge	0.988	0.953	0.970	0.953	0.953	0.953
Macro F1-Score	0.695			0.673		
Overall Accuracy	0.717			0.702		

Table 1: Model A Per-Class Performance: Comparison of Experiment A and Experiment B.

We note that some species such as Ptereleotris and Sponge remain consistently high across both experiments. These likely have very distinct textures that are easier for a scratch-built CNN to extract.

Figure 3 provides a visual comparison of the classification errors for both models. Model A (from scratch) shows [mention a specific trend, e.g., more confusion between similar species], whereas Model B (ResNet-50) demonstrates [mention improvement, e.g., higher diagonal density].

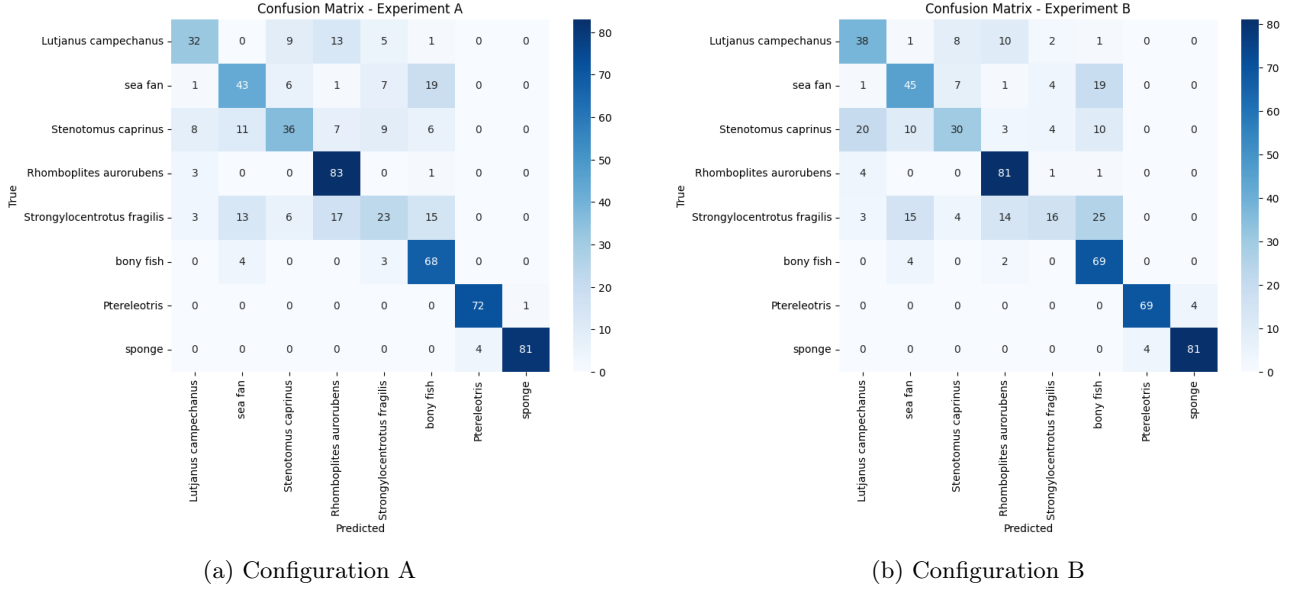


Figure 3: Confusion matrices for both configurations.

6.3 Model B: Transfer Learning Comparison

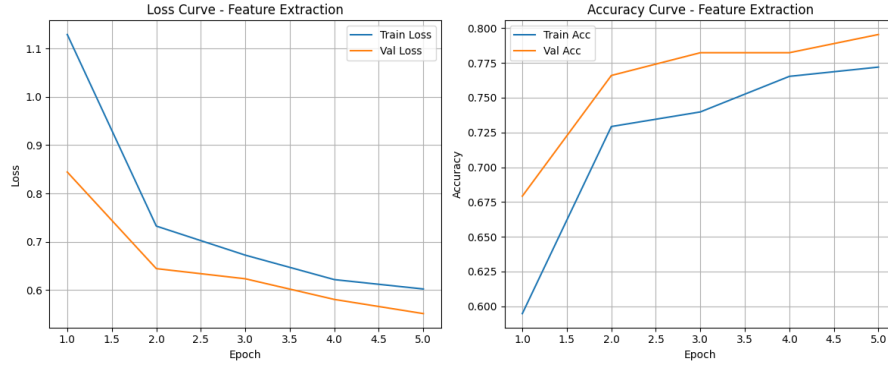
We evaluated the three transfer learning strategies. As expected, full fine-tuning achieved the highest accuracy but required the most computational time.

Strategy	Accuracy	Macro F1	Training Time(s)
Feature Extraction	0.789	0.755	96
Partial Fine-tuning	0.833	0.825	455
Full Fine-tuning	0.864	0.858	1680

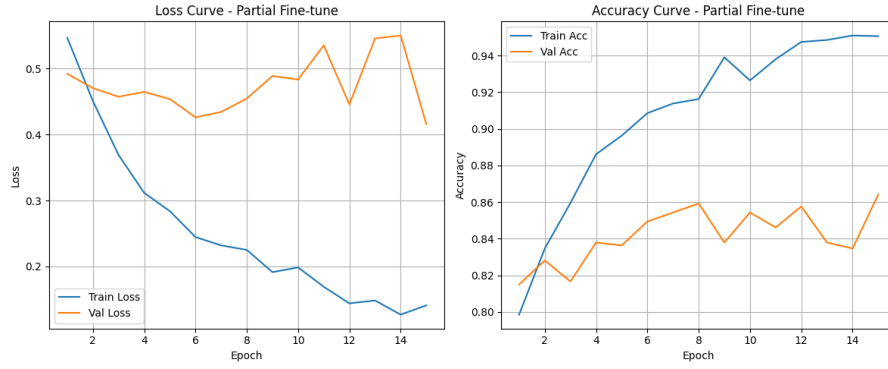
Table 2: Comparison of Transfer Learning Strategies.

6.4 Model B: Training Dynamics Across Stages

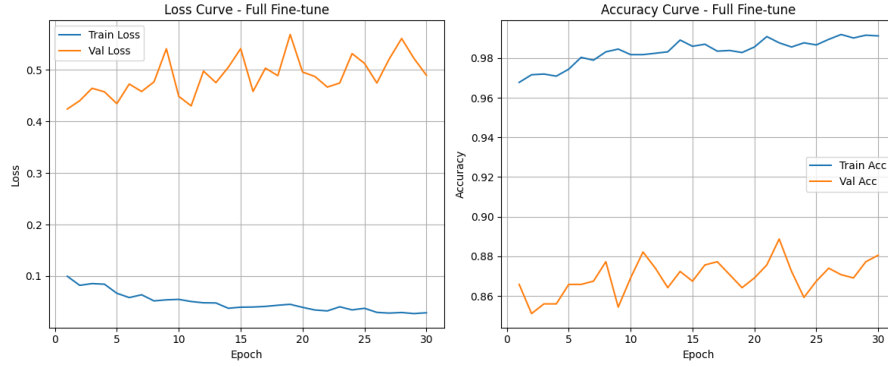
Figure 4 illustrates the three-stage training process for the ResNet-50 model. We observe that as we move from frozen feature extraction to full fine-tuning, the model’s ability to minimize loss on the marine-specific data improves significantly.



(a) Stage 1: Feature Extraction (Backbone Frozen)



(b) Stage 2: Partial Fine-tuning (Layer 4 Unfrozen)



(c) Stage 3: Full Fine-tuning (Entire Backbone Unfrozen)

Figure 4: Comparison of training and validation curves across the three transfer learning strategies. Each stage uses a progressively lower learning rate to maintain architectural stability.

6.5 Model B: Comparative Performance Analysis

The following table summarizes the performance across the three transfer learning strategies. We observe a consistent improvement in both Macro F1 and Overall Accuracy as more layers are unfrozen and specialized to the marine imagery domain.

	Feature Extraction			Partial Fine-tune			Full Fine-tune		
Class Name	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
L. campechanus	0.603	0.733	0.662	0.658	0.833	0.735	0.721	0.817	0.766
Sea fan	0.690	0.779	0.732	0.806	0.649	0.719	0.836	0.727	0.778
S. caprinus	0.615	0.519	0.563	0.710	0.636	0.671	0.775	0.714	0.743
R. aurorubens	0.884	0.966	0.923	0.934	0.977	0.955	0.934	0.977	0.955
S. fragilis	0.694	0.442	0.540	0.685	0.649	0.667	0.764	0.714	0.738
Bony fish	0.756	0.867	0.807	0.817	0.893	0.854	0.833	0.933	0.881
Ptereleotris	1.000	0.973	0.986	1.000	1.000	1.000	1.000	1.000	1.000
Sponge	0.988	0.988	0.988	1.000	1.000	1.000	1.000	1.000	1.000
Macro F1	0.775			0.825			0.858		
Accuracy	0.789			0.833			0.864		

Table 3: Classification metrics for Model B (ResNet-50) under three different training strategies.

6.6 Out-of-Distribution (OOD) Detection Results

The OOD detection performance for both models was evaluated using the dynamically suggested threshold of 0.27 (at 0.95 TPR). Table 4 summarizes the results for the primary Model B configuration.

Metric	Known (In-Distribution)	Unknown (OOD)
Precision	0.803	0.693
Recall	0.293	0.957
Macro F1	0.708	
Overall Accuracy	0.804	

Table 4: Model B OOD Detection Performance Summary.

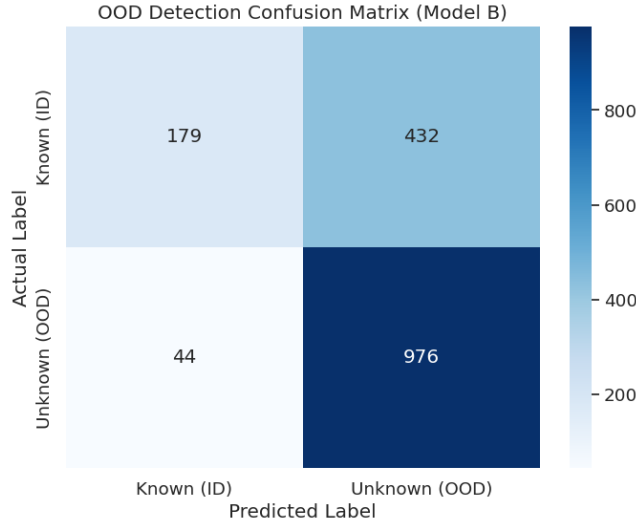


Figure 5: Confusion Matrix for Model B OOD Detection

Although Model B achieved higher overall classification accuracy, Model A (Configuration 2) proved more reliable for OOD detection. Model A provided better separated ID vs OOD confidence levels compared to Model B that was equally confident in both making threshold separation difficult. This is attributed to the fact that larger, pretrained architectures often exhibit overconfidence by forcing novel inputs into a highly-complex learned feature space. In contrast, the more constrained architecture of Model A produced more conservative softmax probabilities for unseen categories, resulting in a cleaner separation between ID and OOD distributions.

A comparison between the two configurations of model A reveals significant trade-offs in their ability to generalize to unseen marine categories. Configuration 2 produced a much cleaner separation in the Maximum Softmax Probability (MSP) distribution. While Configuration 1 often became overconfident on misclassified

samples, the smoother convergence in Configuration 2 forced the model to produce lower confidence scores for inputs that did not align strongly with learned marine features.

A surprising result in our testing was that the known species actually had lower confidence scores than the unknown (OOD) species. While we usually expect a model to be most confident about what it has seen before, this flip makes sense when we look at the specific challenges of the marine dataset. The 10 species we trained on are very similar to each other as an image since they share the same blurry underwater backgrounds. Because they look so much alike, the model often gets confused between two or three correct options, splitting its probability and resulting in a lower overall confidence score (the green bars in Figure 1). On the other hand, the Unknown species likely have very distinct textures or sharp patterns that the model has never seen. These unique features can accidentally set off the model’s internal activation functions much more strongly than the familiar but blurry species it was trained on. This creates a high-confidence false alarm for the unknown species (the red bars). This tells us that in a difficult environment like the ocean, a model being sure of itself can actually be a sign that it is looking at something brand new. By setting our threshold at 0.27, we successfully caught 95.7% of these unknown instances, effectively using the model’s own overconfidence against it to flag potential new species for human review.

7 Conclusion

This project successfully demonstrated the application of deep learning for marine species classification using the FathomNet dataset. Our comparative analysis revealed that while Transfer Learning (Model B) is vastly superior in terms of raw classification accuracy, reaching a peak of 86.4% through full fine-tuning, the scratch-built CNN (Model A) offered unique advantages in terms of transparency and out of distribution reliability. The results showed that increasing the depth of unfreezing in pretrained architectures significantly improves F1-scores for similar species, but this comes at the cost of overconfidence. This was evident in the OOD detection task, where the more constrained Model A provided a cleaner separation between known and unknown instances. A critical takeaway from this study is the discovery of the confidence flip in underwater imagery where the model’s high confidence on unseen species, likely triggered by novel high-contrast textures, can be leveraged as a robust safety flag when calibrated with temperature scaling and dynamic thresholding.

Ultimately, for real-world marine deployments, a hybrid approach is recommended. Utilizing the specialized feature extraction of a fine tuned ResNet for classification, combined with an OOD detection logic tuned to identify high-confidence outliers, would provide the most dependable system for biological monitoring in diverse and unpredictable aquatic environments.